

Table 1. Summary of selected methods and packages under the categories of model-based, algorithm-based and functional clustering approaches for clustering longitudinal data.

Methods	R packages	Description	Criteria to determine the number of clusters	Major strengths	Major weaknesses	Applicable to data scenarios (Figure 2)
Model-based approach						
Group-based trajectory clustering	gbmt (Magrini, 2022)	Finite mixture model based on EM algorithm.	AIC and BIC	Computationally efficient; can include covariates; the timing variable can be modelled explicitly	Assume repeated measurements are independent given the cluster membership	1, 2, 3, 4
Group-based trajectory clustering	flexmix (Leisch, 2004)	Finite mixture model based on EM algorithm.	AIC and BIC	Computationally efficient; can include covariates; the timing variable can be modelled explicitly	Assume repeated measurements are independent given the cluster membership	1, 3
Latent class mixed effect model	lcmm (Proust-Lima et al., 2017)	Latent class mixed effect model based on the finite mixture model and Newton–Raphson-type algorithm.	AIC and BIC	Can incorporate random effect; can model non-linear trajectory; can include covariates; the timing variable can be modelled explicitly; can account for serial correlation	Computationally intensive	1, 2, 3, 4
Latent class mixed effect model	mixAK (Komárek & Komárková, 2014)	Latent class mixed effect model based on finite mixture model and MCMC.	PED	Can incorporate random effect; can include covariates; the timing variable can be modelled explicitly; can incorporate prior knowledge	Computationally intensive	1, 2, 3, 4
Gaussian mixture model	longclust (McNicholas & Murphy, 2010)	Based on a mixture of multivariate t or Gaussian distributions with a Cholesky-decomposed covariance structure.	BIC	Computationally efficient; allows modelling the mixture of multivariate t distribution	Only applicable to balanced data and does not allow missing data; the timing variable cannot be modelled explicitly; does not allow covariates	1, 3
Gaussian mixture model	kmlcov (Subtil et al., 2017)	Clustering using the likelihood as a metric of distance; EM-type algorithm is used.	CLL, AIC and BIC	Can include covariates; the timing variable can be modelled explicitly	Computationally intensive; assume repeated measurements are independent given the cluster membership	1, 3

(Continues)

Table 1 (Continued)

Methods	R packages	Description	Criteria to determine the number of clusters	Major strengths	Major weaknesses	Applicable to data scenarios (Figure 2)
Algorithm-based approach						
K-means clustering	kml (Genolini & Falissard, 2010)	K-means clustering based on Euclidean distance or Manhattan distance.	Ray and Turi criterion, Davies and Bouldin criterion, Calinski and Harabatz criterion, AIC and BIC	Can capture non-linear trajectory with complex shape; no distributional assumptions; computationally efficient	Only applicable to balanced data and does not allow missing data; the timing variable cannot be modelled explicitly; does not allow covariates	1, 3
K-means clustering	kmlshape (Genolini et al., 2016)	K-means using shape-respecting distance.	Not available	Can capture non-linear trajectory with complex shape; no distributional assumptions; computationally efficient	Only applicable to balanced data and does not allow missing data; the timing variable cannot be modelled explicitly; does not allow covariates; no model selection criteria; Does not allow covariates	1, 3
Hierarchical clustering	clusterMLD (Zhou et al., 2023)	Hierarchical clustering based on a dissimilarity metric that quantifies the cost of merging two distinct groups of curves.	Calinski and Harabatz criterion, and gap statistics	Can capture non-linear trajectory with complex shape; computationally efficient		1, 2, 3, 4
Functional data approach						
FPCA with K-means clustering	fdapace (Yao et al., 2005)	First estimate FPC scores then perform K-means clustering on the scores.	Calinski and Harabatz criterion, silhouette statistics, and gap statistics	Can capture non-linear trajectory with complex shape; computationally efficient	Does not allow covariates	1, 2, 3, 4
Model-based clustering for functional data	fdaMocca (Pya Arnqvist et al., 2021)	Each curve is projected onto a finite-dimensional basis and clustering is done on the resulting basis coefficients; an EM-style algorithm is used.	AIC, BIC and Shannon entropy	Can capture non-linear trajectory with complex shape; allow covariates	Computationally intensive	1, 2, 3, 4
Functional subspaces clustering	funFEM (Bouveyron & Jacques, 2015)	Cluster functional data by modelling the curves within a common and discriminative functional subspace.	AIC, BIC and ICL	Can capture non-linear trajectory with complex shape; computationally efficient	Does not allow covariates	1, 3

(Continues)

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<i>Methods</i>	<i>R packages</i>	<i>Description</i>	<i>Criteria to determine the number of clusters</i>	<i>Major strengths</i>	<i>Major weaknesses</i>	<i>Applicable to data scenarios (Figure 2)</i>
Sparse and smooth functional clustering	sasfunclust (Centofanti et al., 2023)	Finite mixture model with a likelihood function penalized with a functional adaptive pairwise fusion penalty and a roughness penalty; an EM-style algorithm is used.	Cross-validation	Can capture non-linear trajectory with complex shape	Computationally intensive	1, 2, 3, 4
Functional clustering using adaptive density peak detection	FADPclust (Ren et al., 2023)	Clustering based on adaptive density peak detection technique in which the density is estimated by functional k-nearest neighbour density estimation.	Calinski and Harabatz criterion, Silhouette statistics, Dunn statistics	Can capture non-linear trajectory with complex shape; computationally efficient	Does not allow covariates	1, 3